Question 1 (40 marks)

You are given the Boolean expression:

$$(A + B)(A + C)(\overline{A} + \overline{B})$$

Perform a complete analysis of this Boolean expression by addressing the following in a single, well-structured response:

(i) **Simplify** the Boolean expression using Boolean algebra theorems. Clearly show each step and justify your simplification. [8 marks]

(ii) **Interpret a truth table** for the original expression and your simplified result. Confirm that both expressions produce identical outputs for all input combinations. [10 marks]

(iii) **Draw and analyze a Karnaugh Map (K-map)** for the original expression using variables A and B, treating C as a "don't care" condition. Use the K-map to verify your simplified result. [7 marks]

(iv) **Interpret the simplified expression** in terms of logic gate functionality. Identify the type of logic gate it represents and describe its behavior. [5 marks]

(v) **Interpret the simplified expression** in a sequential logic context:

Suppose the output of the simplified expression is connected to both the J and K inputs of a JK flip-flop. Interpret the behavior of the flip-flop over time, assuming the initial output is 0. Compare this behavior with a D flip-flop where the complement of the output is fed back into the D input. Discuss how both configurations support toggling and their implications in digital design. [10 marks]

Question 2 (24 marks)

a) Consider a program where 30% of the operations can be executed in parallel, and the rest 70% operations need to be executed sequentially.

i) **Interpret Amdahl's law** to compute the speed up for this program assuming a speed **up factor of 3.** [4 marks]

ii) Interpret the maximum speed up for this program. [4 marks]

b) Reliability is a measure of continuous delivery of the service. The mean time to failure (MTTF) of a system provides a measure of reliability, and can be increased by either improving the quality of the components or designing the system to continue operating in the presence of faulty components. Interpret **three different tactics** that can be used to increase the reliability of the system, based on the previous analysis. [5 Marks]

c) AT&T ESS-1, one of the first computer-controlled telephone exchange was designed for less than two hours of downtime over its lifetime of 40 years.

i) What is the MTTF and MTTR of this system? [4 marks]

ii) Calculate **the percentage of availability** of this system considering that one year has 365 days and one day has 24 hours. [4 marks]

d) The I-format instruction in MIPS has an immediate field of 16 bits. For an I-format addi instruction, these immediate bits are represented using 2's complement representation. Based on these determine the **largest positive value** and the **negative value with the largest magnitude** that can be represented using the bits in the immediate field? [3 marks]

Question 3: (18 Marks)

Use the register and memory values in the table below for the following questions. Assume a 32-bit machine with standard addressing modes as discussed in Hennessy & Patterson. Each question starts from the values in the table independently.

Register	Value	Memory Address	Value
R1	20	20	40
R2	32	32	12
R3	48	48	24
R4	64	64	16

(i) What are the values of **R1**, **R2**, and **R3** after executing the instruction:

add R3, R1, R2 *(Hint: R3 ← R1 + R2)* [6 marks]

(ii) What are the values of **R1** and **R3** after executing the instruction:

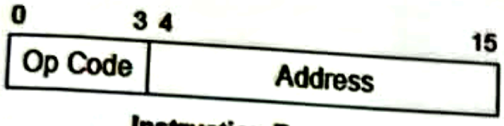
load R3, 4(R1) *(Hint: R3 ← Memory[R1 + 4])* [6 marks]

(iii) What are the values of **R2** and **R4** after executing the instruction:

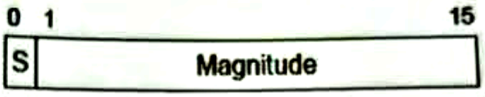
addi R2, R4, -8 *(Hint: R2 ← R4 + (-8))* [6 marks]

Question 4: (18 Marks)

A hypothetical machine has the following characteristics



Instruction Format



Integer Format

Program Counter (PC)
Instruction Register (IR)
Accumulator (AC)

Internal CPU Registers

0001 = Load AC from Memory
0010 = Store AC to Memory
0101 = Add to AC from Memory
0011 = Load AC from I/O
0111 = Store AC to I/O

Partial List of Opcodes

Address	Contents
400	1540
401	5541
402	2541
540	0002
541	0003

Address	Contents
PC	400
AC	000A
IR	0000

ıe 12-bit address identifies a particular I/O device. Given the respective memory :ations with its respective content, illustrate the program execution steps when the ɔgram Counter (PC) is loaded with a value of 400 upon code start-up (Hint: Draw 6 ɔs). [18 marks]

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